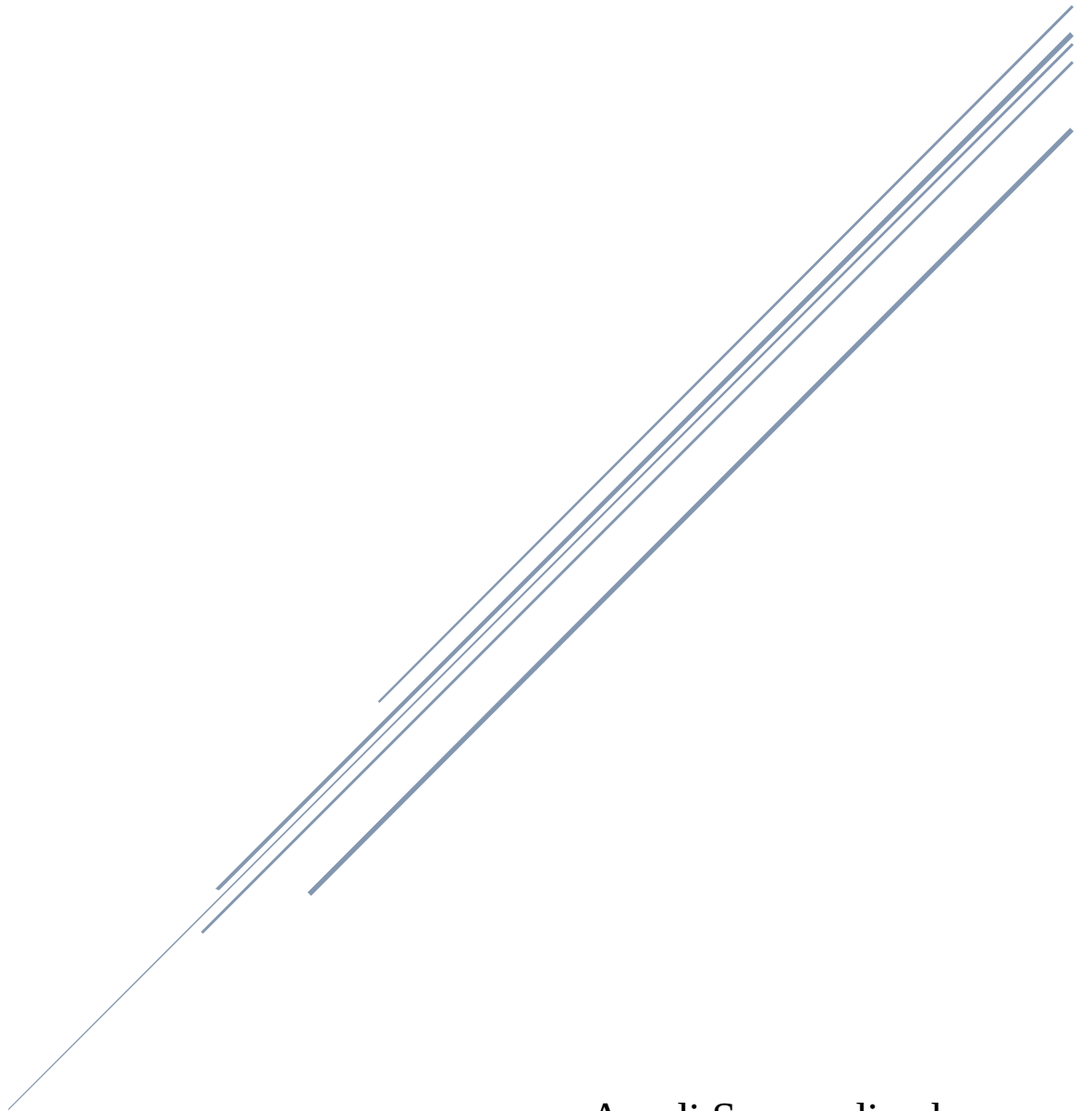


Basic Encryption and Decryption Using the Caesar Cipher

DecodeLabs Cyber Security Internship – Project 2



Anudi Samaradiwakara
Cybersecurity

Basic Encryption and Decryption Using the Caesar Cipher

DecodeLabs Cyber Security Internship – Project 2

1. Introduction

Data security has become one of the most important aspects of modern technology. Organizations and individuals continuously exchange information through digital platforms, making the protection of sensitive information essential. Encryption is one of the most widely used techniques for protecting data from unauthorized access.

This project focuses on implementing a basic encryption and decryption system using the Caesar cipher algorithm. The purpose of the project is to understand the fundamental concepts of cryptography and develop practical experience in protecting information through mathematical and logical operations.

2. Project Objective

The primary objective of this project is to create a system capable of converting plain text into encrypted text and then restoring it to its original form.

The project aims to achieve the following goals:

- Understand the basic principles of cryptography.
- Learn how encryption and decryption processes work.
- Strengthen programming and logical thinking skills.
- Build a secure and user-friendly application.
- Explore the importance of data confidentiality.

3. Problem Statement

As digital communication continues to increase, information is constantly exposed to security threats such as unauthorized access, data theft, and data manipulation. Therefore, it is essential to protect information before transmitting it across networks.

The Caesar cipher provides a simple but effective method for understanding how encryption techniques protect data. Although it is not suitable for highly sensitive information, it serves as an excellent foundation for learning modern cryptographic techniques.

4. Scope of the Project

The scope of the project includes the following activities:

- Accepting user input.
- Encrypting the entered text.
- Decrypting the encrypted text.
- Displaying the original, encrypted, and decrypted outputs.
- Allowing users to select a custom shift value.

5. Technologies Used

The following technologies were used during the development process:

Technology

- Python
- Flask
- HTML
- CSS
- JavaScript

6. Methodology

The project follows a straightforward process consisting of several stages.

Step 1 – User Input

The user enters the text that needs to be encrypted.

Step 2 – Encryption Process

Each character in the text is shifted by a fixed number of positions in the alphabet.

Step 3 – Displaying the Encrypted Text

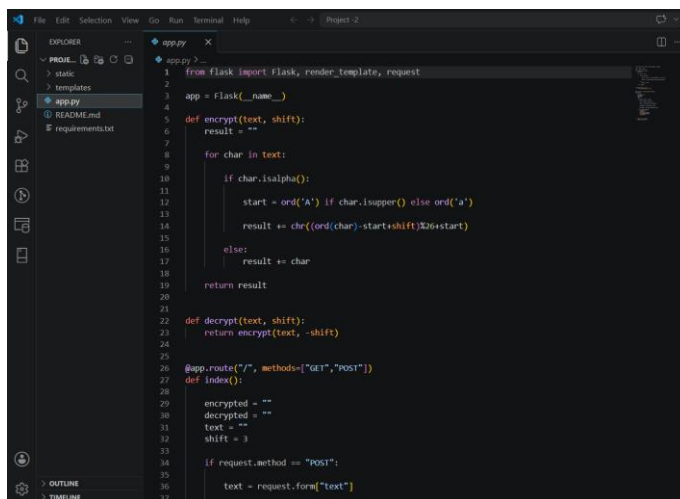
The encrypted output is displayed to the user.

Step 4 – Decryption Process

The encrypted text is shifted back to its original position.

Step 5 – Displaying the Original Message

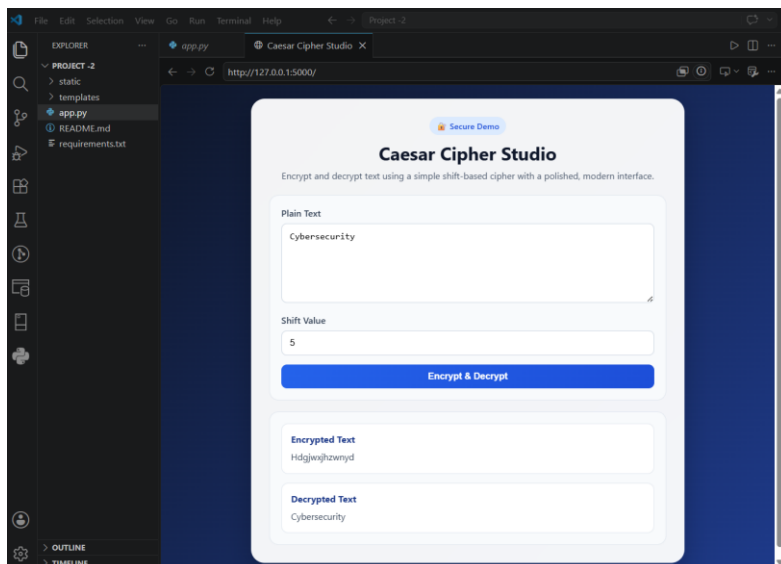
The system displays the decrypted output.



```
1 from flask import Flask, render_template, request
2
3 app = Flask(__name__)
4
5 def encrypt(text, shift):
6     result = ""
7
8     for char in text:
9         if char.isalpha():
10             start = ord('A') if char.isupper() else ord('a')
11             result += chr((ord(char) - start + shift) % 26 + start)
12         else:
13             result += char
14     return result
15
16 def decrypt(text, shift):
17     return encrypt(text, -shift)
18
19 @app.route("/", methods=["GET", "POST"])
20 def index():
21     encrypted = ""
22     decrypted = ""
23     text = ""
24     shift = 3
25
26     if request.method == "POST":
27         text = request.form["text"]
```

```
File Edit Selection View Go Run Terminal Help Project-2
EXPLORER
PROJECT-2
static
templates
index.html
app.py
README.md
requirements.txt
index.html
1 <doctype html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Caesar Cipher Studio</title>
7   <link rel="stylesheet" href="{url_for('static', filename='style.css')}">
8 </head>
9 <body>
10   <div class="page-shell">
11     <div class="container">
12       <div class="hero">
13         <div class="badge"> Secure Demo</div>
14         <h1>Caesar Cipher Studio</h1>
15         <p>Encrypt and decrypt text using a simple shift-based cipher with a polished, modern interface.</p>
16       </div>
17       <form method="POST" class="form-card">
18         <label for="text">Plain Text</label>
19         <input id="text" type="text" placeholder="Enter text to encrypt..." required/>
20
21         <label for="shift">Shift Value</label>
22         <input id="shift" type="number" name="shift" value="{(shift)}" min="1" max="26"/>
23
24         <button type="submit">Encrypt & Decrypt</button>
25       </form>
26
27       <div if encrypted %>
28         <div class="result-card">
29           <div class="result-box">
30             <h2>Encrypted Text</h2>
31             <p>{{encrypted}}</p>
32           </div>
33           <div class="result-box">
34             <h2>Decrypted Text</h2>
35             <p>{{decrypted}}</p>
36           </div>
37     </div>
```

```
File Edit Selection View Go Run Terminal Help Project-2
EXPLORER
PROJECT-2
static
templates
index.html
app.py
README.md
requirements.txt
style.css
1 {
2   box-sizing: border-box;
3 }
4
5 body {
6   margin: 0;
7   font-family: "Segoe UI", Tahoma, Geneva, Verdana, sans-serif;
8   background: linear-gradient(135deg, #00172a, #00172a, #00172a);
9   color: #fff372;
10   min-height: 100vh;
11 }
12
13 .page-shell {
14   min-height: 100vh;
15   display: flex;
16   justify-content: center;
17   align-items: center;
18   padding: 24px;
19 }
20
21 .container {
22   width: 100%;
23   max-width: 768px;
24   background: #00172a;
25   padding: 32px;
26   border-radius: 24px;
27   box-shadow: 0 20px 40px #00172a;
28   backdrop-filter: blur(10px);
29 }
30
31 .hero {
32   text-align: center;
33   margin-bottom: 24px;
34 }
35
36 .badge {
37   display: inline-block;
```



7. Algorithm Description

The Caesar cipher is one of the oldest and simplest encryption techniques in history.

The algorithm performs the following operations:

- Read the input text.
- Select a shift value.
- Replace each letter with another letter based on the selected shift value.
- Preserve spaces, symbols, and numbers.
- Generate the encrypted message.
- Reverse the process to obtain the original message.

8. Features of the System

The application provides the following features:

- Encryption functionality.
- Decryption functionality.
- User-friendly interface.
- Responsive design.
- Support for uppercase and lowercase characters.
- Support for numbers and symbols.
- Adjustable shift value.
- Fast processing speed.

9. Advantages

The main advantages of this project are listed below:

- Easy to understand and implement.
- Improves problem-solving skills.
- Introduces the principles of cryptography.
- Strengthens programming knowledge.
- Provides practical cybersecurity experience.

10. Limitations

Despite its advantages, the Caesar cipher has several limitations:

- It provides only basic security.
- It is vulnerable to brute-force attacks.
- It uses a limited number of possible shift values.
- It is not suitable for highly confidential information.

11. Future Improvements

Several enhancements can be added to improve the application in the future:

- Implementation of the Vigenère cipher.
- Advanced encryption standards (AES).
- File encryption capabilities.
- Multiple encryption methods.
- User authentication mechanisms.
- Encryption history storage.

12. Conclusion

This project provided valuable experience in understanding the fundamental principles of cybersecurity and cryptography. The implementation of the Caesar cipher demonstrated how information can be transformed into an unreadable format and later restored to its original form.

Through this project, a strong foundation was established for exploring more advanced encryption algorithms and security techniques in the future.